

Track 1: Full Stack Development

June 2026

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MR ROKESH TECHNOLOGY — Training & Internship Programs

Skill path: HTML/CSS → JavaScript/TypeScript → React → Node/Express → Spring Boot → REST APIs → Auth → Payments → Deploy

Related: [Master Syllabus](#) | [Duration Framework](#) | [Project Progression](#)

Track Overview

Who this is for: Aspiring web developers, B.Tech/BCA/MCA students, career switchers targeting frontend, backend, or full-stack roles.

Prerequisites: Basic computer literacy; no prior coding required for 2-week/1-month tiers. 6-month and 9-month tiers benefit from completing 1-month foundation or equivalent self-study.

Tools & lab environment: - VS Code, Git, GitHub, Node.js (LTS), npm/yarn - React, Express, MongoDB/PostgreSQL - Postman, Chrome DevTools - Render/Vercel/Netlify (free tier deployment) - Optional (6mo+): Docker, Spring Boot, Razorpay/Stripe test keys

Certification targets (9-month): Portfolio-ready projects; optional AWS Cloud Practitioner alignment via [Track 8](#)

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Awareness; needs mentorship	1 deployed full-stack mini app
1 Month	80–100	Junior basics	Portfolio site + Blog CMS (MERN mini)
3 Months	250–300	Internship-ready	3–4 projects + E-commerce/Job Portal capstone
6 Months	500–600	Entry–mid job-ready	5–6 projects + production capstone
9 Months	750–900	Industry-ready	8–10 projects + 2 advanced capstones (SaaS + RBAC dashboard)

Weekly Rhythm (All Durations)

Day	Activity
Mon–Thu	Concept (30%) + live coding lab (70%)
Fri	Mini-project submission
Sat	Code review + doubt clearing
Sun	Revision + GitHub README documentation

2-Week Crash Course

Mini Project: Deployed full-stack notes app | **Level:** L1

Assessment: Quiz (20%) + Lab checkpoints (40%) + Project demo (30%) + README (10%)

Week 1

Day	Topics	Lab Objective	Deliverable	Self-Check
D1	HTML5 semantic tags, document structure	Build semantic page skeleton	<code>index.html</code> with header, nav, main, footer	Name 5 semantic tags and their purpose
D2	CSS Flexbox, Grid, responsive design	Style landing page for mobile + desktop	Responsive personal landing page	Explain Flexbox vs Grid use cases
D3	JavaScript: variables, loops, functions, DOM	Manipulate DOM elements dynamically	Interactive elements on landing page	Write a function that filters an array
D4	Events, fetch API, JSON	Fetch data from public API and display	API-powered section on page	What is the difference between sync and async?
D5	React: JSX, components, props	Convert static page section to React component	React app with 2+ components	What are props and why are they read-only?
D6	React: useState, useEffect hooks	Build stateful UI with side effects	Component with dynamic state	When does useEffect run?
D7	React portfolio site (integration)	Combine week's topics into portfolio	React portfolio site on GitHub	Demo portfolio to mentor

Week 1 submission checklist: - ☐ Landing page responsive at 375px and 1280px widths - ☐ To-Do or interactive feature working - ☐ React portfolio pushed to GitHub with README

Week 2

Day	Topics	Lab Objective	Deliverable	Self-Check
D1	Node.js setup, Express basics, routing	Create Express server with 3 routes	Running Express server	What is middleware?
D2	REST API design, CRUD operations	Build notes API (GET, POST, PUT, DELETE)	REST API for notes app	Name all HTTP methods and their uses
D3	Connect frontend to API, error handling	React app consumes notes API	Full-stack notes UI + API	How do you handle 404/500 errors?
D4	MongoDB/MySQL basics, schema design	Create notes collection/table	Database schema + seed data	SQL vs NoSQL — when to use each?
D5	API + DB integration (Mongoose or mysql2)	Persist notes to database	Database-driven notes app	Explain CRUD at the database layer
D6	JWT auth basics (register, login)	Add auth to notes API	Protected routes with JWT	What goes in a JWT payload?
D7	Deploy to Render/Vercel, env variables	Deploy frontend + backend live	Live deployed full-stack app	Walk through deployment steps

Week 2 submission checklist: - [] API CRUD works via Postman and UI - [] Auth flow tested (register → login → access protected route) - [] Live URL in README; no secrets in repo

1-Month Foundation

Projects: Portfolio website, Weather app, Blog frontend, Blog CMS (MERN mini) | **Levels:** L1–L2

Week	Topics	Lab Focus	Project Milestone
W1	HTML/CSS/JS fundamentals, Git/GitHub, responsive design, accessibility basics	Daily HTML/CSS/JS labs; Git workflow	Portfolio website live on GitHub Pages
W2	ES6+ (arrow functions, destructuring, modules), async/await, fetch, React fundamentals (components, hooks)	Async API calls; React component architecture	Weather app with OpenWeather API
W3	React Router, controlled forms, lifting state, basic context	Multi-page React app with form validation	Blog frontend (3+ routes)
W4	Node/Express, REST conventions, MongoDB, Mongoose, JWT auth, deployment	Full-stack integration lab	Blog CMS (MERN mini) deployed

Weekly lab duration: 3–4 hrs/day Mon–Thu (70% hands-on)

W4 Capstone acceptance criteria (Blog CMS): - ☐ Create, read, update, delete blog posts via UI - ☐ User registration and login with JWT - ☐ Author can only edit own posts - ☐ Deployed with environment variables for DB URI and JWT secret - ☐ README with screenshots and live URL

3-Month Job Ready

Capstone: E-commerce OR Job Portal (MERN) | **Level:** L3

Month	Topics	Projects	Milestones
M1	Frontend mastery: HTML5, CSS3 (Flexbox/Grid), JS ES6+, React, Tailwind CSS, responsive + accessible UI	Portfolio (polished), To-Do app, Weather app	3 frontend projects on GitHub
M2	Backend: Node.js, Express, REST API design, PostgreSQL + MongoDB, Mongoose/Prisma, JWT auth, error handling, validation	Task manager API, Blog CMS (enhanced)	API documented in Postman; auth tested
M3	Full-stack integration, Spring Boot intro, Razorpay/Stripe test payments, CI/CD basics (GitHub Actions), capstone	Capstone: E-commerce or Job Portal	Production deploy; payment flow OR job application flow

M3 Capstone acceptance criteria: - [] User auth (register, login, profile) - [] Core domain CRUD (products/jobs + cart/applications) - [] Admin role with elevated permissions - [] Payment integration (test mode) OR application tracking workflow - [] GitHub Actions CI runs tests on push - [] 10-minute demo + architecture diagram

6-Month Professional

Capstone: E-commerce / Social Media Clone / Job Portal | **Level:** L4

Aligned with Imperial College Bootcamp + Ada Developers Academy + TechKnowledgeHub

Month	Topics	Projects
M1	HTML5, CSS3, JS ES6+, DOM, responsive design, WCAG accessibility	Portfolio + Calculator
M2	Advanced JS (closures, prototypes), async patterns, React (hooks, context, React Router)	To-Do + Weather + Blog UI
M3	React advanced (custom hooks, performance), Tailwind CSS, Jest + React Testing Library, SPA deployment	Full React SPA with routing + tests
M4	Node.js, Express, REST APIs, middleware, error handling, multer file upload, API versioning	REST API + Auth backend
M5	MongoDB + PostgreSQL, Mongoose, SQL joins, indexing, query optimization, transactions	Database-driven app (dual DB or migration exercise)
M6	Spring Boot intro, JWT/OAuth2, Razorpay/Stripe, Docker basics, capstone + career prep	Capstone: E-commerce / Social Media Clone / Job Portal

M6 Capstone acceptance criteria (L4 rubric): - ☐ All L3 criteria met - ☐ Payment gateway OR social feed with real-time updates - ☐ Admin dashboard for content/user management - ☐ Minimum 5 automated tests (Jest) - ☐ Docker Compose for local dev environment - ☐ CI/CD pipeline deploys on merge to main - ☐ Case study document submitted

9-Month Advanced

Capstones: (1) SaaS product (2) Enterprise dashboard with RBAC | **Level:** L5

Additional modules beyond 6-month track: - TypeScript across frontend and backend - Next.js (SSR/SSG/App Router) - Microservices basics (API gateway pattern) - Redis caching, WebSockets (real-time chat module) - GraphQL intro (Apollo Server + Client) - AWS/Vercel deployment pipelines - Open-source contribution sprint (1 merged PR or meaningful contribution) - Mock interviews + system design basics (URL shortener, chat system)

Phase	Months	Focus
Foundation	M1–M6	Complete 6-month curriculum
Specialization	M7	TypeScript + Next.js migration of capstone
Advanced backend	M8	Microservices, Redis, WebSockets, GraphQL
Capstones	M8–M9	SaaS product + Enterprise RBAC dashboard

Capstone 1 — SaaS Product acceptance criteria: - ☐ Multi-tenant or subscription model (Stripe billing) - ☐ Next.js App Router with SSR/SSG where appropriate - ☐ TypeScript strict mode enabled - ☐ Redis caching for frequent queries - ☐ Monitoring/logging configured

Capstone 2 — Enterprise RBAC Dashboard acceptance criteria: - ☐ Role-based access control (admin, manager, viewer) - ☐ Real-time updates via WebSockets - ☐ GraphQL or REST API with comprehensive docs - ☐ System design document with trade-offs - ☐ Load tested with basic k6 or Artillery script

Assessment Rubric (Track-Specific)

Component	Weight	Criteria
Weekly Quiz	20%	HTML/CSS/JS/React/Node concepts; code reading
Lab Completion	40%	All checkpoints submitted; code runs; follows style guide
Project	30%	Functionality, architecture, demo quality
Portfolio	10%	GitHub README, live URL, commit history, case studies

Track-specific project grading emphasis: - Frontend weeks: UI responsiveness, accessibility, component structure - Backend weeks: API design, error handling, security (no exposed secrets) - Full-stack weeks: End-to-end flow, deployment, documentation

Recommended Resources

Resource	Used For
MDN Web Docs	HTML, CSS, JavaScript reference
React Documentation	React hooks, components, patterns
Node.js Documentation	Node/Express backend
Ada Developers Academy Curriculum 2024	Full Stack structure
Imperial College Bootcamp Syllabus	Project-based learning model
TechKnowledgeHub 6-Month Curricula	Pacing and project sequencing

Appendix: Mentor Notes

Common pitfalls: - Students skip Git commits until project end — enforce daily commits from Week 1 - Confusion between client-side and server-side validation — demonstrate both explicitly - Environment variables not used; secrets committed to Git — run secret scan in code review - React state mutation bugs — teach immutable update patterns early

Hardware requirements: - Minimum 8 GB RAM, 256 GB storage - Stable internet for deployment labs and npm installs

Optional stretch goals: - Add dark mode toggle to portfolio - Implement infinite scroll on blog - Add OAuth (Google login) in 3-month+ tiers - Migrate MERN app to TypeScript in 9-month tier

Cross-track references: - Database depth: [Track 2 — Database Technologies](#) - DSA interview prep: [Track 3 — Programming Languages](#) - Deployment/CI/CD depth: [Track 7 — Cloud & DevOps](#) - Career modules: [Track 14 — Career Development](#)